

Demo Board For RV1106

RV1106G AI IPC(38Demo) V1.1

Reference Design Main Functions Introduction	
Power	4 DCDC + 4 LDO (or 3 DCDC + 4 LDO)
RAM	SPI FLASH
Interface	SDMMC/SDIO/MIPI CSI/LVDS RX/VI_CIF/ VI_BT656/VI_BT1120/I2S/USB/ADC

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Project:		IPC-P01-Z02	
File:		00.Cover Page	
Date:	Monday, February 27, 2023	Rev:	V1.1
Designed by:	Linau	Reviewed by:	Default
		Sheet:	1 of 17

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Index and Notes

Note

NOTE 1:

Component parameter description

1. DNP stands for component not mounted temporarily
2. If Value or option is DNP, which means the area is reserved without being mounted

NOTE 2:

Please use our recommended components to avoid too many changes.

For more informations about the second source,please refer to our AVL.

Generate Bill of Materials

Header:

Item\tPart\tDescription\tPCB Footprint\tReference\tQuantity\tOption

Combined property string:

{Item}\t{Value}\t{Description}\t{PCB Footprint}\t{Reference}\t{Quantity}\t{Option}

Graphic Description



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Project:		IPC-P01-Z02	
File:		01.Index and Notes	
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Revision History

Version	Date	Author	Change Note	Approved
V1.0	2022.04.21	Linxu	First edition;	
V1.1	2022.05.27	Linxu	1.Update RV1106G symbol; 2.Change netname of SDMMC0 to SDMMC; 3.Change C1000 to 4.7uF. 4.Change R2005 to 47K. 5.Change DOVDD1V8 to DOVDD 1V8. 6.Add R1006 and connect R8I00 to CODEC_AVSS 7.Swap R8100/R8101 and C8101/C8102.	

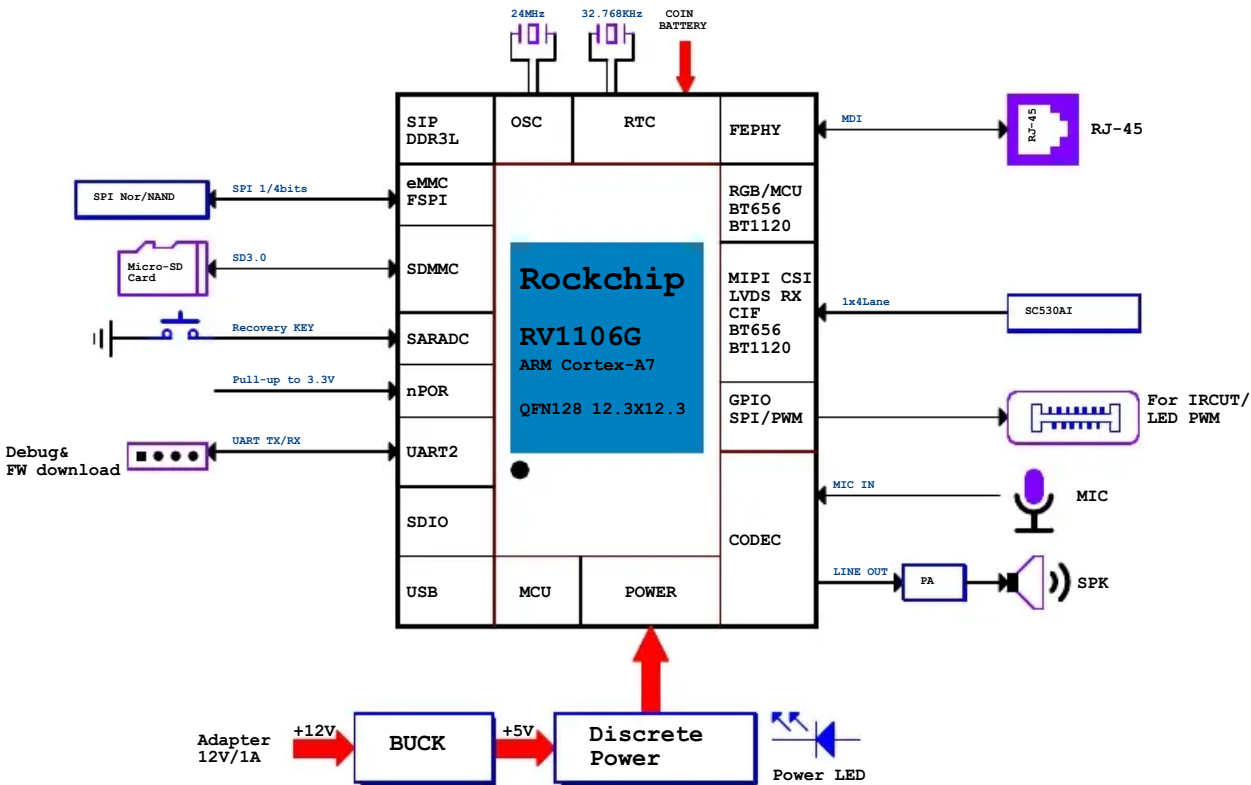
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Project:		IPC-P01-Z02	
File:		02.Revision History	
Date:	Monday, February 27, 2023	Rev:	V1.1
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Block Diagram



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Project: IPC-P01-202

File: 03.Block Diagram

Date: Monday, February 27, 2023

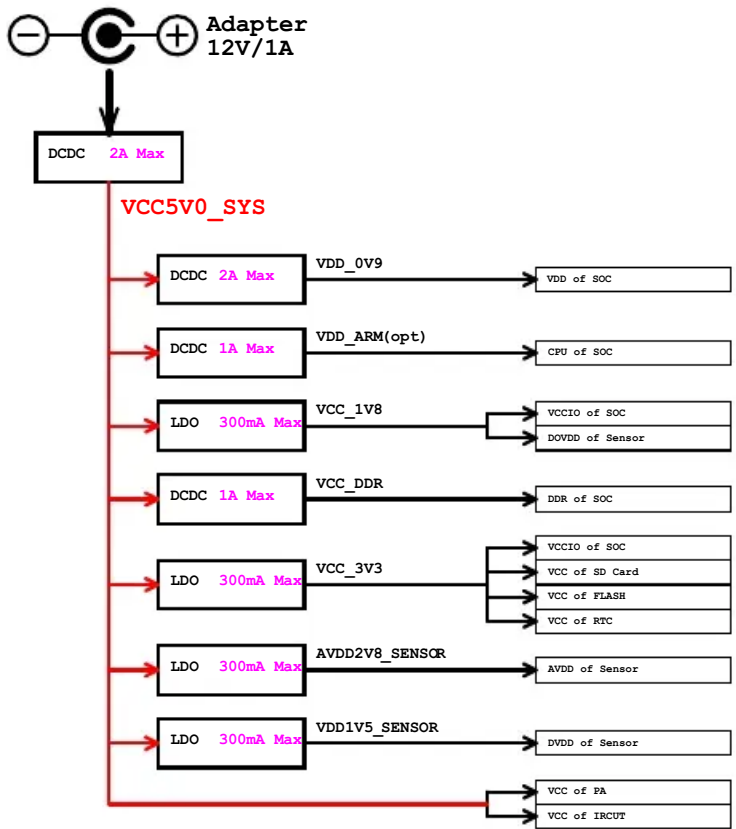
Design: Lian

Review: Lian

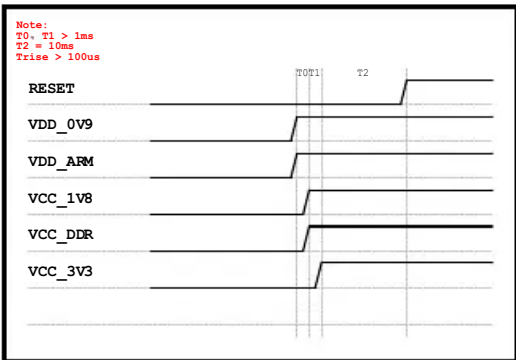
Draw: Lian

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Power Diagram and Sequence



Power-on Sequence				
Power Name	PMIC Channel	Time Slot	Default voltage	Peak Current
VCC5V0_SYS	DC-DC BUCK		5.0V	1.5A
VDD_0V9	DC-DC BUCK	Slot: 1	0.93V	1.6A
VDD_ARM	DC-DC BUCK	Slot: 1	0.93V	0.5A
VCC_1V8	LDO	Slot: 2	1.8V	0.1A
VCC_DDR	DC-DC BUCK	Slot: 2	1.35V	0.5A
VCC_3V3	LDO	Slot: 3	3.3V	0.2A
RESET	Finally, nPOR RESET 10ms after VCC_3V3 is ready			



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File: 04.Power Diagram and Sequence			
Date:	Monday, February 27, 2023	Rev:	V1.1
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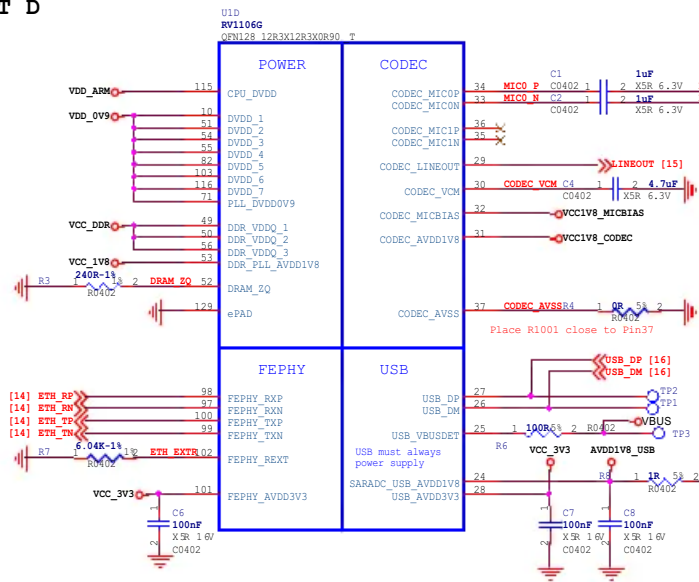
IO Power Domain Map

IO Domain	IO Group	Support of IO Voltage		Default Actual assigned IO Domain Voltage			Remark
		1.8V	3.3V	Net Name of Power Supply	Power Source	Voltage	
PMU	Group GPIO0_A		✓	VCC_3V3		3.3V	
VCCIO1	Group GPIO1_AB		✓	VCC_3V3		3.3V	
VCCIO2	Group GPIO4_C	✓		VCC_1V8		1.8V	
VCCIO3	Group GPIO4_AB	✓	✓	VCCIO_FLASH		1.8/3.3V	
VCCIO4	Group GPIO3_A	✓	✓	VCCIO_SD		1.8/3.3V	
VCCIO5	Group GPIO2_AB	✓	✓	VCC_3V3		3.3V	
VCCIO6	Group GPIO1_CD	✓	✓	VCC_3V3		3.3V	
VCCIO7	Group GPIO3_BCD	✓		VCC_1V8		1.8V	

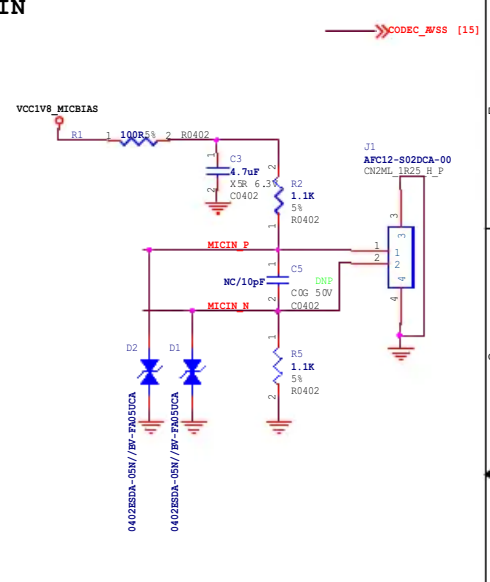
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Project:		IPC-P01-Z02	
File:		06.IO Power Domain Map	
Date:	Monday, February 27, 2023	Rev:	V1.1
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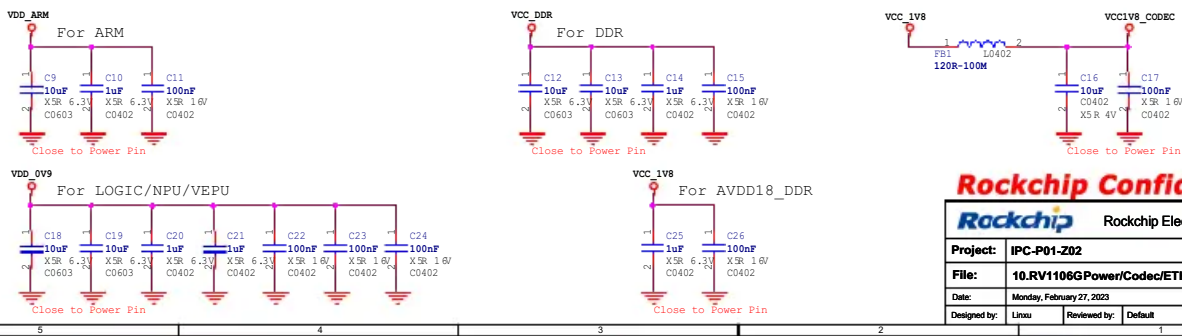
PART D



MIC IN



POWER



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Project: IPC-P01-Z02

File: 10.RV1106GPower/Codec/ETH/USB

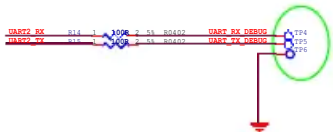
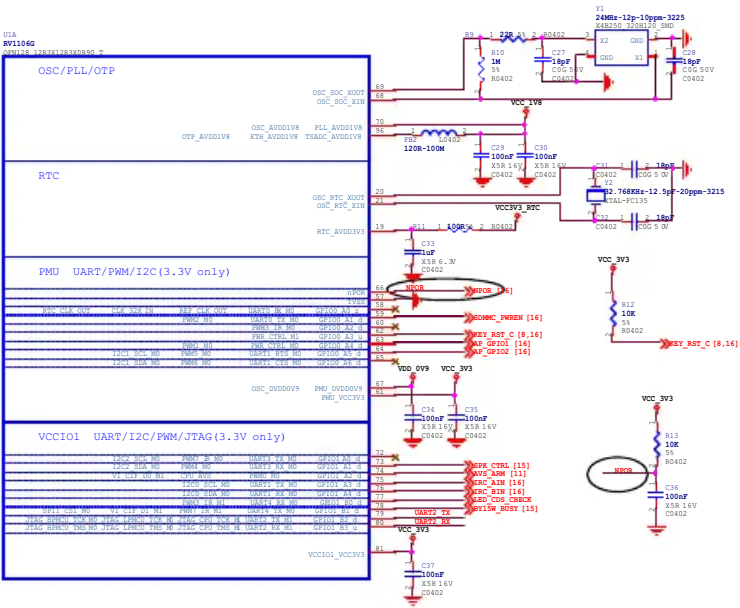
Date: Monday, February 27, 2023

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Rev: V1.1

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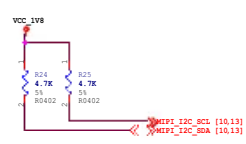
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Project: IPC-P01-202

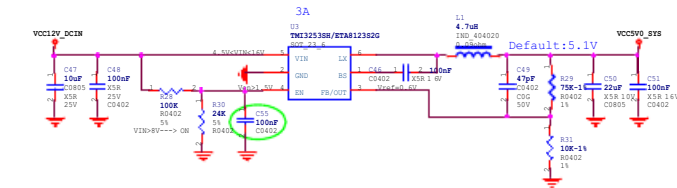
File: 11.RV1106G OSC/RTC/PMU/VCCIO

Date: Monday, February 27, 2023 Rev: V1.1

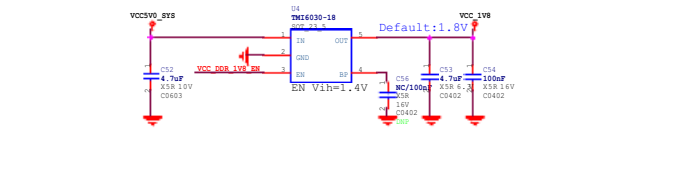
Designby: Leno Revisionby: Dabul Sheet: 5 of 17



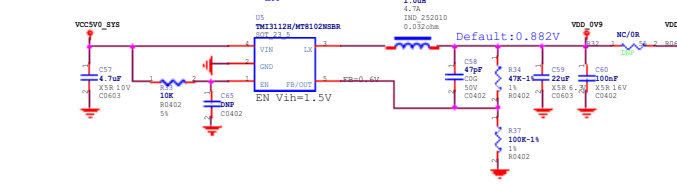
12V---->VCC5V0_SYS



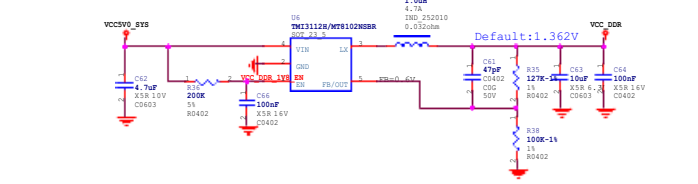
VCC5V0_SYS---->VCC_1V8
Setp 2



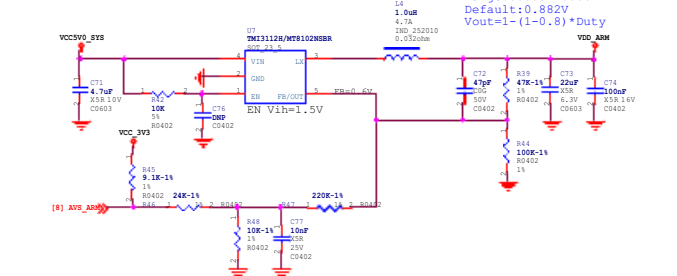
VCC5V0_SYS---->VDD_0V9
Setp 1



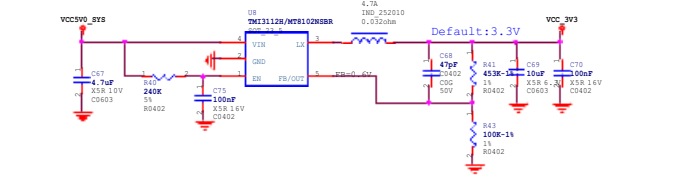
VCC5V0_SYS---->VCC_DDR
Setp 2



VCC5V0_SYS---->VDD_ARM
Setp 1



VCC_5V---->VCC_3V3
Setp 3

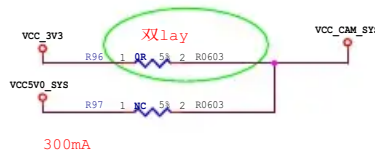
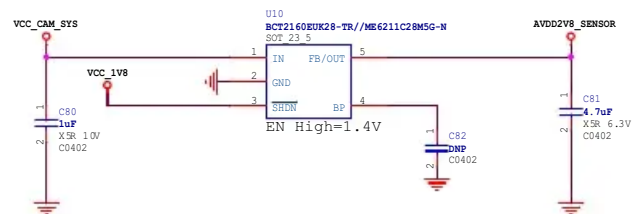
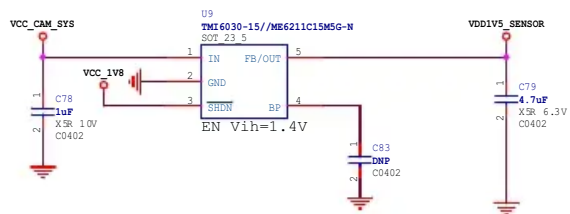


RTC Power



VCC5V0_SYS--->VDD1V5_SENSOR

VCC5V0_SYS--->AVDD2V8_SENSOR

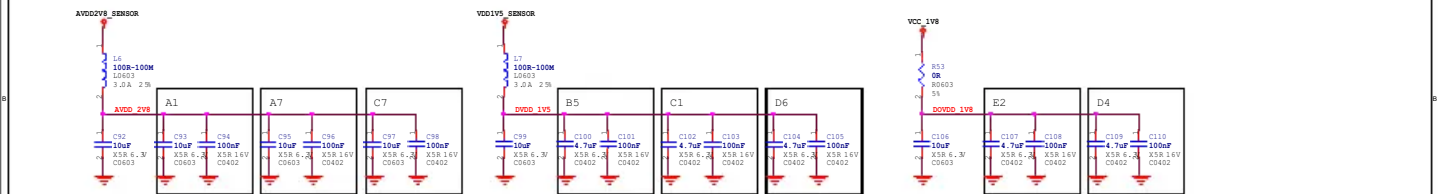
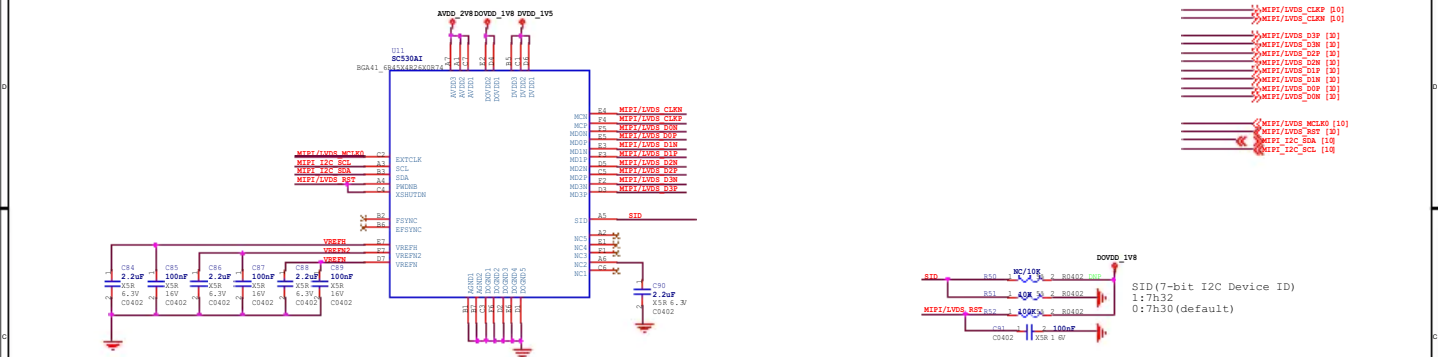


Note:
Default power-on timing:
All three power on at the same time.
Or DOVDD(VCC_1V8)-->DVDD(VCC_1V5)-->AVDD(VCC_2V8)

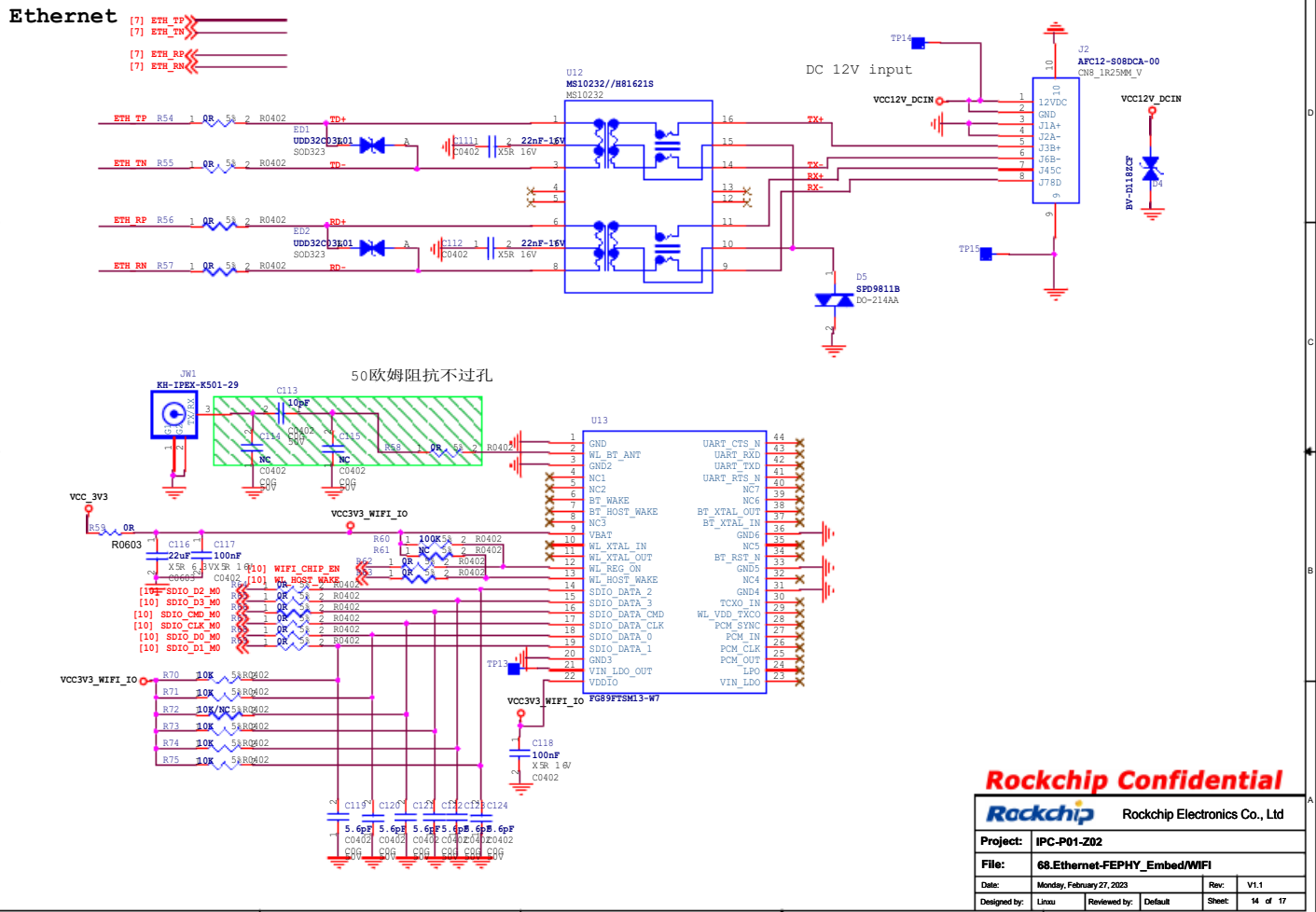
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Project:	IPC-P01-Z02		
File:	45.VI-Camera Power		
Date:	Monday, February 27, 2023		Rev: V1.1
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Sensor



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Project:	IPC-P01-Z02
File:	47.VI-Camera-SC530AI
Date:	Monday, February 27, 2023
Designby:	Linu
Reviewedby:	Default
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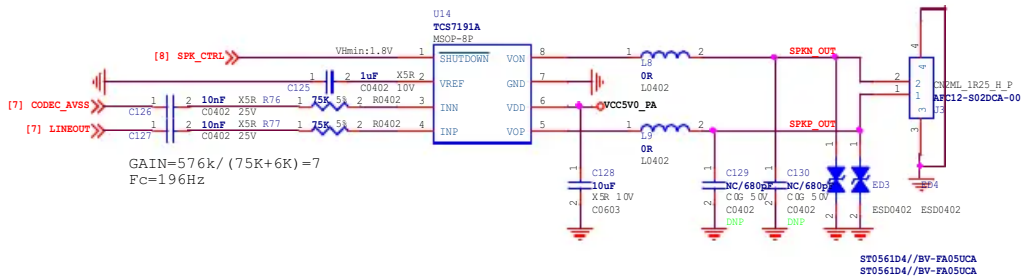
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Project:		IPC-P01-Z02	
File:		68.Ethernet-FEPHY_Embed/WFI	
Date:	Monday, February 27, 2023	Rev:	V1.1
Designed by:	Linux	Reviewed by:	Default
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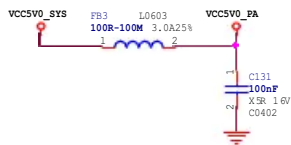
Speaker Out

NOTE:If there is no EMI problem, the ferrite beads can be replaced by 0ohm resistors;

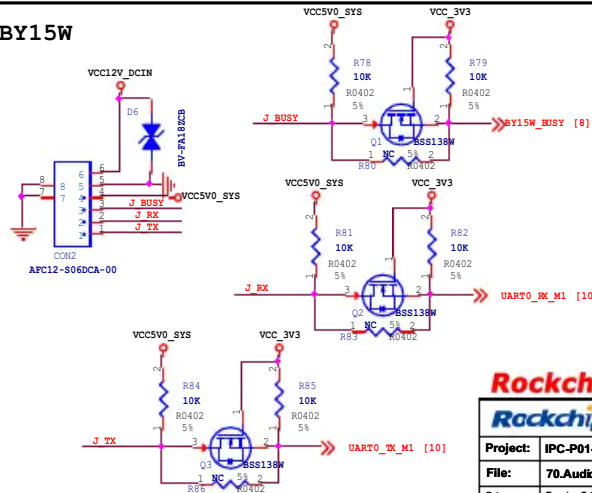
NOTE:CODEC_AVSS and LINEOUT are routed according to the difference rule;



Speaker power



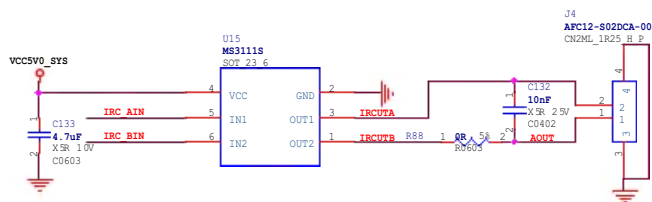
BY15W



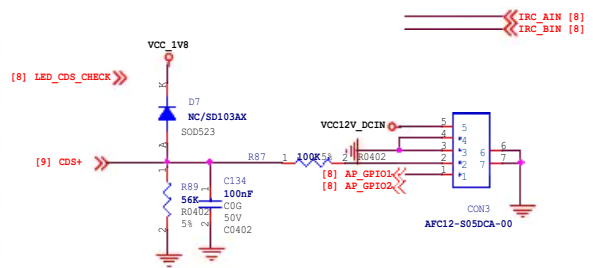
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Project: IPC-P01-Z02			
File: 70.Audio Port/BY15W			
Date:	Tuesday, February 28, 2023	Rev:	V1.1
Designed by:	Linuo	Reviewed by:	Default
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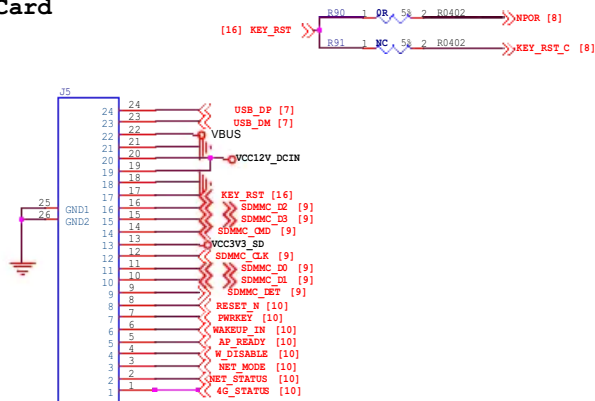
IR-CUT Drive



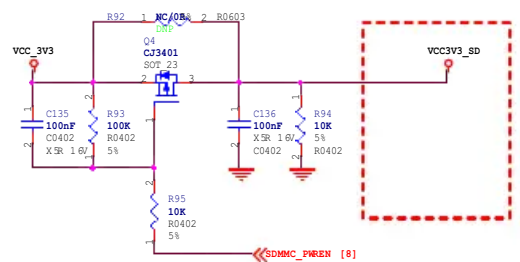
CDS&LED



Micro-SD Card



Card Power

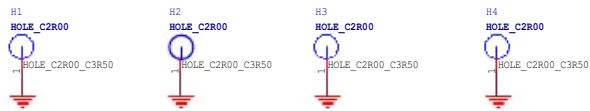


SDMMC0_PWREN=L VCC3V3_SD=3.3V(Default)
SDMMC0_PWREN=H VCC3V3_SD=0V

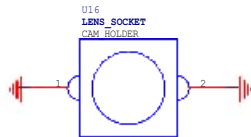
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Project:		IPC-P01-Z02	
File:		82.Micro-SD Card/IRCUT/CDS	
Date:		Monday, February 27, 2023	Rev: V1.1
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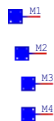
HOLE



LENS



MARK



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Project:	IPC-P01-Z02		
File:	99.MARK/HOLE		
Date:	Monday, February 27, 2023	Rev:	V1.1
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